

Capstone Proposal

Inventory Monitoring at Distribution Centers

Domain Background

In distribution Centers like in that of Amazon/Flipkart, robots are often used to move objects between different warehouse locations as part of their day-to-day operations. Different objects depending on their size are clubbed together in a single bin for efficient movement of objects. So, a single bin can carry one, two three, four or five objects (or even more). This project aims to build a Computer Vision Model that can be used to count number of objects in each bin. A system like this can be used to track inventory and make sure that delivery consignments have the correct number of items.

Problem Statement

You are provided with different image files containing different pictures of bins. These bins can carry from 1-5 objects. Problem involves counting number of objects in each of the bin images irrespective of type, shape and packaging of the objects. Such problems will be encountered by any distribution center using such techniques to move objects from one place to another. So, problem is replicable, quantifiable and measurable.

Solution Statement

We need to have an automated system that can count number of objects present in bin as shown in an image file. This is basically a Computer Vision problem and can be solved in two ways: through object localization and detection and through classification. Our choice of solution is to use classification approach as such solutions are easy to train and implement. We will be using pre-trained CNN models for this purpose. We will freeze the CNN model, attach a new classification head on top of it and train only the classification head part of it using the provided dataset. Once the model is fine-tuned, it can be evaluated and then deployed to end-point where it can be used for inference. System will input an image file to the end-point. Model deployed at endpoint will perform classification and send number of objects present in the image back to calling system. Such a solution can be used in different distribution centers using similar technique to move around objects. We can also assess the error rate of model post deployment.

Datasets and Inputs:

We are going to use Amazon Bin Image Dataset which has 500,000 bin images containing one or more objects. Due to resource constraints, we will be using subset of dataset provided with project.

The subset contains following classes of image files followed by number of images belonging to each class:

Class 1: 1228

Class 2: 2299

Class 3: 2666

Class 4: 2373

Class 5: 1875

We will divide the dataset subset into training, validation and test datasets in following proportions stratified by image classes:

Train – 90%

Validation – 5%

Test – 5%

Details about full dataset can be found at [Amazon Bin Image Dataset - Registry of Open Data on AWS](#)
This project involves counting of objects present in a bin. This dataset contains 500,000 images of bin containing 1 or more objects. Hence the dataset is very much suitable given the problem/project context.

Benchmark Model

Vision Transformer is State of the Art model in the field of image classification. This is SOTA model which is pre-trained on millions of images as part of which it learns to recognize innumerable shapes, sizes, lines, curves, textures. This pre-trained information contained in model when fine-tuned on a specific dataset provides speedy and efficient training with high level of accuracy even though the original model might not have seen the new images ever in their lifetime. This process is called transfer learning. More information with respect to model accuracy can be found at [An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale | Papers With Code](#)

Evaluation Metrics

We will be using accuracy to assess the model performance.

Accuracy = (number of images correctly classified)/(total number of images in dataset)

Due to resource constraints, we are using a subset of dataset and hence the metrics like recall/precision might not be appropriate here and may not present true picture with respect to full dataset.

Project Design

Project flow looks like as under. We will try to emulate a production setup as much as possible.

- 1) Download the dataset and divide into training/validation and test dataset and then transfer it to S3 location.
- 2) We will be using resnet50 and efficientnet_b0 pretrained models.
- 3) These pretrained models will be fine-tuned on given dataset using defined hyperparameter values.
- 4) The model which performs best in step 3 will further undergo hyperparameter tuning.
- 5) Best model obtained in step4 will then be retrained using best hyperparameters with sagemaker debugger and profiler option.
- 6) Model performing best in step 3 will also undergo multi-instance model training.
- 7) Out of Steps 5 and 6, whichever model performs best, will further be chosen for model deployment

- 8) Model deployment will be done using single instance/multi-model endpoint/multiple production variants setup. We will also setup endpoint rescaling.
- 9) Once the entire setup is done, we will create an end to end model pipeline and perform all the steps using pipeline starting from model training point onwards.
- 10) We will then, create a lambda function to invoke deployed model endpoint and integrate it with AWS API gateway
- 11) We will perform inference on deployed model using public URL provided by API gateway using gradio interface.